

CDP (Cisco Discovery Protocol)

CDP (Cisco Discovery Protocol) is a Layer 2 proprietary protocol developed by Cisco that allows Cisco devices to discover and share information with directly connected Cisco devices.

Features of CDP:

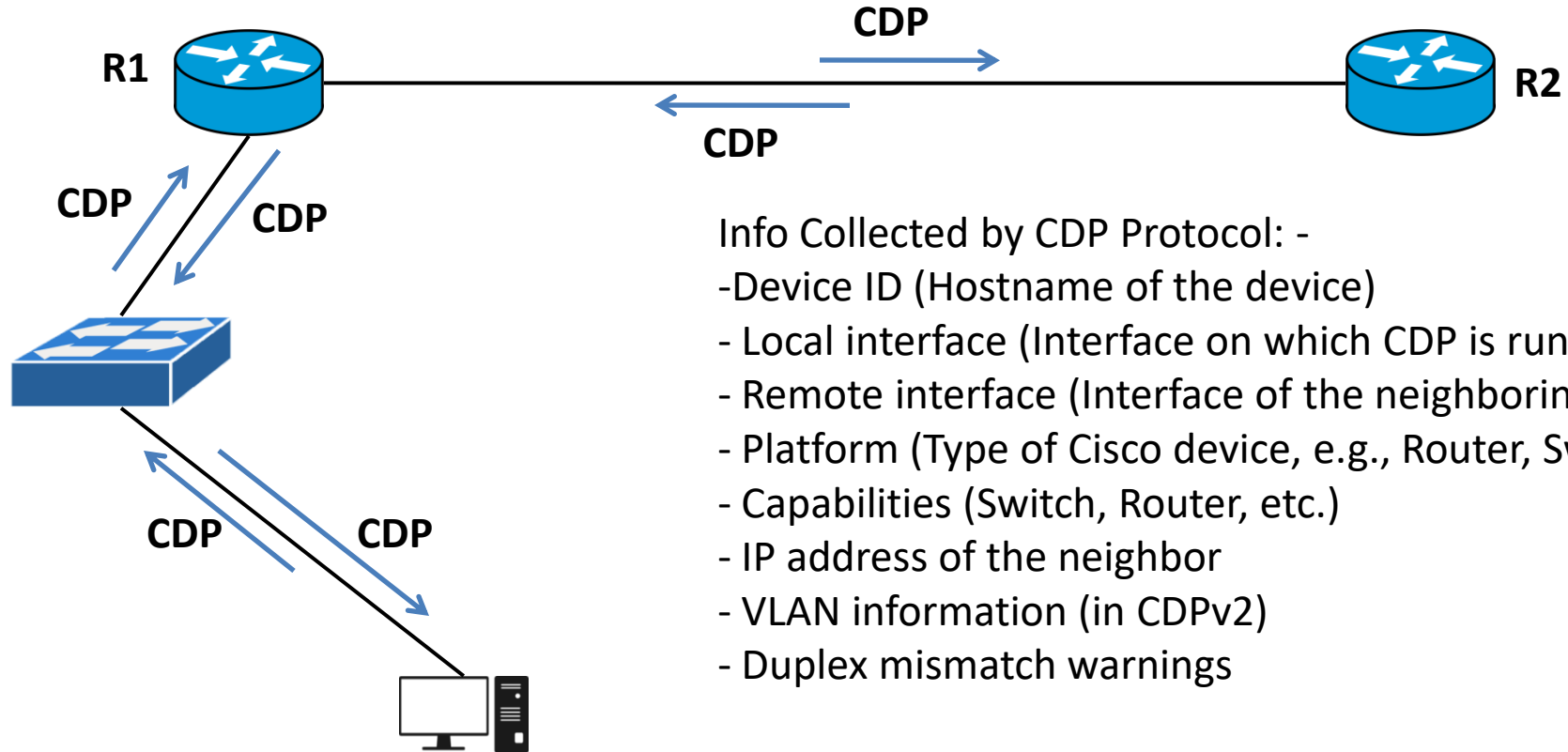
1. It is a Layer 2 Protocol:

- Operates at the **Data Link Layer** (Layer 2) of the OSI model.
- Works on **point-to-point** connections.

2. Device Discovery:

- CDP allows a Cisco device to **discover other Cisco devices** (routers, switches, etc.) that are directly connected.
- Helps in network topology mapping and troubleshooting.

CDP protocol:



Info Collected by CDP Protocol: -

- Device ID (Hostname of the device)
- Local interface (Interface on which CDP is running)
- Remote interface (Interface of the neighboring device)
- Platform (Type of Cisco device, e.g., Router, Switch)
- Capabilities (Switch, Router, etc.)
- IP address of the neighbor
- VLAN information (in CDPv2)
- Duplex mismatch warnings

Key Features of CDP:

3. Periodic Updates:

- Devices **send CDP advertisements** periodically (default: **60 seconds**).
- Advertisement messages contain device information.

4. Protocol Independent:

- CDP runs **independently of Layer 3 protocols** like IP or IPX.

5. Supports Multiple Protocols:

- Can provide details about **IP address, VLAN, and other protocol-related information**.

6. CDP Versions:

- **CDPv1**: Basic information exchange.
- **CDPv2**: Adds additional features like **VTP domain details, native VLAN mismatch detection, and more**.

CDP Information Shared: -

CDP provides the following details about a neighboring Cisco device:

- Device ID (Hostname of the device)
- Local interface (Interface on which CDP is running)
- Remote interface (Interface of the neighboring device)
- Platform (Type of Cisco device, e.g., Router, Switch)
- Capabilities (Switch, Router, etc.)
- IP address of the neighbor
- VLAN information (in CDPv2)
- Duplex mismatch warnings

CDP Timers: -

CDP (Cisco Discovery Protocol) uses two main timers to control how often information is sent and how long neighbor information is retained.

1. CDP Timer

- Controls how often **CDP advertisements** are sent.
- Default value: **60 seconds**.

2. CDP Holdtime

- Specifies how long a device **retains neighbor information** if no new CDP packets are received.
- Default value: **180 seconds**.
- If the holdtime expires before receiving a new advertisement, the neighbor entry is removed.

CDP timer Commands:-

Command to modify the CDP timer:

Router(config)# cdp timer <seconds>

Router(config)# cdp timer 30

Command to modify the CDP holdtime:

Router(config)# cdp holdtime <seconds>

Router(config)# cdp holdtime 120

Verifying CDP Timers:

Router# show cdp

CDP Commands:

CDP must be **enabled globally** and **on an interface** to function.

1. Enable or Disable CDP Globally

- Enable CDP globally

Router(config)# cdp run

- Disable CDP globally

Router(config)# no cdp run

2. Enable or Disable CDP on an Interface

- Enable CDP on an interface:

Router(config-if)# cdp enable

- Disable CDP on an interface

Router(config-if)# no cdp enable

Show CDP Information:

- Display basic details of neighbors

Router# show cdp neighbors

- Detailed info, including IP address

Router# show cdp neighbors detail

- Show information for all neighbors:

Router# show cdp entry

- Shows interfaces running CDP

Router# show cdp interface

Displays CDP statistics (packets sent/received)

Router# show cdp traffic

Basic Neighbor Table:-

Command for CDP Neighbors

Router# show cdp neighbors

Device ID	Local Intf	Holdtime	Capability	Platform	Port ID
Switch-1	Gig0/1	180	S I	WS-C2960X	Gig0/24
Router-2	Gig0/2	120	R S I	ISR4331	Gig0/1

Device ID: Neighbor's hostname

Local Intf: Interface on the local device where CDP is running

Holdtime: Time before the CDP entry expires

Capability: S (Switch), R (Router), I (IGMP), etc.

Platform: Model of the neighboring device

Port ID: Neighbor's interface

CDP vs LLDP:

Feature	CDP	LLDP
Vendor-Specific	Cisco Only	Multi-Vendor (IEEE 802.1AB)
Layer	Layer 2	Layer 2
Default Timer	60 sec	30 sec
Default Holdtime	180 sec	120 sec
Protocol Support	Cisco devices	All devices
VLAN Information	Yes	Yes

Conditions in Which CDP Disabled:

CDP can expose network details to attackers if not controlled properly.

- **Disable CDP on External Interfaces (e.g., Internet-facing ports)**
- **Disable CDP on Trunk Links** where it's unnecessary.

Command to Disable CDP:

- Disable globally:

Router(config)# no cdp run

- Disable on a specific interface

Router(config-if)# no cdp enable